

Su Tüketimi Tahmin Modelleri – Sonuç Raporu

Pencere=12 ay | 10 seed ortalaması | Train:96 / Val:24 / Test:12 örnek

1. Her Modelin Seed Bazlı Test Sonuçları

ROCKET (num_kernels=1000)

Seed	RMSE	MAE	MAPE (%)
1	4489.07	3792.67	19.8129
2	3698.26	3161.81	16.5801
3	4073.01	3157.89	16.8713
4	4683.54	3828.25	20.2248
5	4666.87	3953.30	20.7488
6	4730.59	4010.98	20.9931
7	3785.59	2910.55	15.4969
8	4299.25	3460.84	18.3795
9	4987.48	4160.27	21.9151
10	4522.19	3727.26	19.6682
Ortalama	4393.58	3616.38	19.0691

LSTM-FCN (lstm_size=16, batch_size=16)

Seed	RMSE	MAE	MAPE (%)
1	3635.53	3127.49	15.9776
2	2755.20	2218.85	11.3489
3	3977.37	3131.85	15.1897
4	3727.83	2939.16	13.8272
5	3465.76	2789.85	13.5389
6	4951.12	4216.89	21.1979
7	3525.31	2912.17	15.1362
8	6518.59	4638.05	21.6045
9	3784.56	3211.22	16.6044
10	3796.58	2403.26	12.7300
Ortalama	4013.78	3158.88	15.7155

InceptionTime (batch_size=8, n_filters=32, depth=3)

Seed	RMSE	MAE	MAPE (%)
1	6312.06	5416.65	25.5897
2	4067.21	3429.36	16.0188
3	4334.73	3838.05	18.3107
4	2066.51	1651.55	8.3403
5	2418.65	2079.61	9.9727
6	4377.92	3817.78	19.9571

Seed	RMSE	MAE	MAPE (%)
7	2648.05	2281.76	11.7021
8	3251.10	2730.43	14.3161
9	2899.89	2337.51	11.9903
10	2603.71	2178.54	10.5053
Ortalama	3497.98	2976.13	14.6703

Random Forest

Seed	RMSE	MAE	MAPE (%)
1	2131.08	1702.43	8.8942
2	2114.22	1693.22	8.8885
3	1917.77	1488.52	7.7789
4	2111.78	1707.02	8.8860
5	2166.57	1726.37	9.0126
6	2069.25	1590.14	8.3372
7	2145.90	1830.77	9.4253
8	2120.63	1775.71	9.1765
9	2205.81	1828.02	9.4609
10	2048.61	1625.05	8.4517
Ortalama	2103.16	1696.73	8.8312

XGBoost

Seed	RMSE	MAE	MAPE (%)
1	2386.13	1838.03	9.7052
2	2724.11	1941.44	10.3865
3	2550.66	1880.13	9.9715
4	2676.28	1865.16	9.9203
5	2317.61	1739.00	9.1646
6	2431.57	1825.36	9.5989
7	2733.79	1951.19	10.4055
8	2605.21	1935.85	10.1932
9	2437.50	1880.96	9.9583
10	2341.88	1837.26	9.6506
Ortalama	2520.47	1869.44	9.8955

DecisionTree

Seed	RMSE	MAE	MAPE (%)
1	4832.40	3712.64	18.1724
2	2313.20	2057.59	10.3273
3	2556.17	2037.92	10.5431
4	2556.17	2037.92	10.5431

Seed	RMSE	MAE	MAPE (%)
5	3543.08	2968.62	15.4620
6	3696.29	2910.60	15.1033
7	3414.08	2890.07	14.3846
8	1865.03	1547.32	7.8510
9	2556.17	2037.92	10.5431
10	3430.25	2867.32	14.4369
Ortalama	3076.29	2506.79	12.7367

2. Validasyon Performansları (RMSE sınırlaması)

Sıra	Model	RMSE	MAE	MAPE (%)	MASE	RMSSE
1	ROCKET	4346.54	3466.27	17.8973	1.411	1.361
2	Random Forest	5249.16	4794.72	22.2090	1.952	1.644
3	XGBoost	5563.95	5057.48	22.9556	2.059	1.742
4	DecisionTree	5885.71	5157.60	23.8802	2.100	1.843
5	InceptionTime	6078.97	5345.89	24.6168	2.176	1.904
6	LSTM-FCN	6120.29	5218.77	24.8074	2.125	1.917
7	ARIMA	7644.48	7105.14	30.9984	2.893	2.394
8	Naive	8761.95	8145.67	35.1917	3.316	2.744
9	Seasonal Naive	9731.58	9023.25	39.8621	3.673	3.048
10	SARIMA	10555.95	9752.55	42.7840	3.970	3.306

3. Test Performansları (RMSE sınırlaması)

Sıra	Model	RMSE	MAE	MAPE (%)	MASE	RMSSE	Süre (s)
1	Random Forest	2103.16	1696.73	8.8312	0.659	0.586	1.07
2	XGBoost	2520.47	1869.44	9.8955	0.726	0.703	0.31
3	DecisionTree	3076.29	2506.79	12.7367	0.974	0.858	0.74
4	InceptionTime	3497.98	2976.13	14.6703	1.156	0.975	73.83
5	LSTM-FCN	4013.78	3158.88	15.7155	1.227	1.119	52.30
6	Seasonal Naive	4177.05	3652.00	18.8602	1.419	1.164	0.00
7	ROCKET	4393.58	3616.38	19.0691	1.405	1.225	29.77
8	ARIMA	4422.96	3943.78	20.5477	1.532	1.233	0.53
9	Naive	5941.53	5642.33	28.9245	2.192	1.656	0.00
10	SARIMA	10109.39	9458.49	47.9470	3.675	2.818	0.06

4. Multi-Seed Özeti (Std Dahil)

Model	Mean RMSE	Std RMSE	Mean MAE	Mean MAPE (%)	N
ROCKET	4393.58	401.93	3616.38	19.0691	10
LSTM-FCN	4013.78	979.87	3158.88	15.7155	10

Model	Mean RMSE	Std RMSE	Mean MAE	Mean MAPE (%)	N
InceptionTime	3497.98	1219.52	2976.13	14.6703	10
Random Forest	2103.16	74.94	1696.73	8.8312	10
XGBoost	2520.47	150.08	1869.44	9.8955	10
DecisionTree	3076.29	823.81	2506.79	12.7367	10